Microservices in Edge and Cloud Computing for Safety in Intelligent Transportation Systems

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***Abstract*—In the last years there has been a strong effort in the development of Intelligent Transportation System (ITS)-based solutions, leading to an important change in the way that drivers and other road users become aware of the surroundings. The development of Cooperative-ITS, which utilises direct wireless short-range connections, is integrating cellular networks as well (4G and 5G), allowing the growing use of the road users smartphones to provide real-time information about Vulnerable Road Users (VRUs), like pedestrians and cyclists. Such increase is becoming a serious concern, since every VRU is likely to have one smartphone, which may lead to scalability and latency issues. This work presents an approach for a microservices-based application, targeting the always critical VRU safety use-case, in a multi-site scenario, using real road infrastructure Multi-Access Edge Computing (MEC) and mobile network provider cloud computing. The main novelty of this work is the multiple approaches on the deployment of the required microservices, and several scenarios that have been tested, and the investigation of the best approach to minimize the service-level latency of the safety application. The results show the potential of microservices distribution through the edge and cloud, with a strong impact on improving the efficiency of ITS. Depending on the services' location, the latency of the VRU and vehicle's notification is deeply affected, but using a federated scenario we are able to keep the VRU's notification delay around the 200 ms, with better results being achieved if a closer mobile network provider cloud platform is used. The results present a noticeable advancement in the development of more scalable and operational solutions that work on improving ITS, with a focus on microservices and edge computing to minimize the delay of critical applications.**

***Index Terms*—Intelligent Transportation Systems, Vulnerable Road Users, Micro-services, Multi-Access Edge Computing, Cloud, Vehicular Safety.**

I. Introduction

In the last couple of years there have been technological advances in Intelligent Transportation Systems (ITSs) targeting the deployment of new solutions for different modes of transportation and traffic management, *e.g.* enabling users to be better informed about road conditions [1]. Within the ITS main goals, the safety of VRUs is paramount. There has been an evolution towards the use of road users' smartphones to warn them about traffic accidents, as well as to avoid potential accidents [2]. However, such use cases tend to leverage services deployed at a single central cloud [3]. With a growing access of devices into the central cloud, the network load and data transmission delay can be greatly increased, which may induce latency and reliability problems in VRU safety services [4].

Multi-Access Edge Computing (MEC) is a possible solution to avoid the communication with the cloud by migrating services to the edge of the infrastructure. The idea of creating applications aimed at the safety of VRUs is made possible with MEC-based architectures that create computing capabilities close to mobile nodes, enabling applications to run on the edge with less core services intervention, making the deployment of time-critical applications less prone to network congestion, and more capable of fulfilling latency and reliability requirements.

One of the key challenges within MEC based solutions is the placement of the deployment of the services in order to use the best computational nodes [5]. The definition of the best node depends on the scenario, *e.g.* in autonomous driving services the best nodes are the ones with the highest computational capabilities, while critical ITS safety services require the minimization of latency. The best node might even be on a different stakeholder infrastructure, *e.g.* infrastructure from an Mobile Network Operator (MNO). Here, MEC federation can be considered in order to allow the seamless usage by services of all infrastructures [6]. In fact, ITS, Vehicle to Everything (V2X) use cases and associated services are defined by European Telecommunications Standards Institute (ETSI) as a set of possible use cases [7], with their impact on latency analysed, for example, in the work in [8].

The placement of the services shall be transparent to developers of ITS services through an abstraction of all available edge and cloud resources [9]. However, to maximize the benefits of considering a federated MEC, the design of the services shall avoid monolithic solutions and favor decomposable, flexible microservices-based architectures, where each small decoupled component of the service can be deployed throughout the several available computational nodes [10].

Solutions for collision avoidance ITS services involving VRU and vehicles have been the subject of much research but not always considering MEC, federation of MECs, microsservices architecture paradigm nor the usage of cellular infrastructure. The work in [11] investigated the potential for using real crash and near-crash data for the evaluation of collision avoidance algorithms. It defines the concept of time

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to collision, frequently used to measure how likely a potential collision is. The work in [12] proposed a Vehicle to Pedestrian (V2P) framework that combines collision prediction services and wireless communication technologies to avoid accidents between pedestrians and vehicles, especially in cases where the driver does not have visual contact with the pedestrian due to obstacles. The conclusion from this study is that there is a strong room for improvement in the prediction of the VRUs' next move/next short path, and this work proposed a prediction framework based on V2P communication that can provide enhanced road safety to the VRU. However, one limitation of the proposed framework is that only vehicles receive notifications of possible risks, not including the VRUs.

Regarding complete end-to-end collision avoidance solutions, the work in [13] explored the implementation of a V2P system framework where road users communicate with each other via a cloud platform using unidirectional communications based on a publish-subscribe paradigm, through cellular technologies, namely 4G and 5G-NR. The proposed system architecture consists of a V2P client application, an edge cloud, and a V2P server application. The V2P server application processes the ITS messages from the edge servers and predicts collision risks, broadcasting the warning messages to relevant road users. This solution is mostly centralized and does not consider the advantages of leveraging MEC devices or the cellular infrastructure.

The work in [14] proposed a system that is composed of a user-side application, which periodically sends messages containing the VRU position, speed, and orientation, and a MEC-based application that can process the geographical location of the VRU, predicts a possible collision and warns the parties involved when needed. To use regular cellphones and integrate them into the ITS scheme, the work in [15] implemented an architecture capable of enabling interoperability between Intelligent Transport Systems Generation 5 (ITS-G5) and Wi-Fi. Taking into consideration that the network load can increase significantly using MEC solutions, the work in [16] aimed to reduce the load on the downlink side by running filtering and prioritization on the MEC application. Filtering the beacons, e.g. beacons generated by vehicle nodes are only sent to pedestrian nodes, already reduces the number of beacons disseminated, lowering the congestion. This work concluded that prioritization strategies can improve the latency in MEC-based VRU systems.

The work in [17] proposed, implemented and tested a system that focuses on improving the safety of Vulnerable Road Users (VRUs) within a smart city environment by using a multi-sensor system such as radars, video cameras, and communication messages, using ITS-G5, 4G and 5G, all fused together to predict and warn about potential collisions. This is the most complete work from the ones found in the literature, since it integrates both communication and sensing, and an integration between the edge and cloud. However, it is still not optimized in its implementation approach, since it does not leverage the MEC nor the latency improvements that may arise from leveraging the cellular infrastructure.

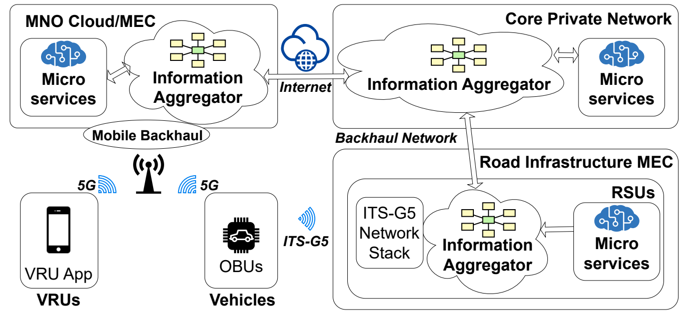


Fig. 1: General architecture of an ITS microservices-based system with multi-site operation.

The main contribution of this paper is to propose, deploy and test a new architecture for collision detection involving vehicles and VRUs, making use of the MEC road infrastructure of a real testbed leveraging ITS-G5 and 5G communications, and cloud capacity for service deployment with routing optimization in the cellular network. The work follows a microservices-based paradigm, and several strategies, regarding the location of the several services, will be evaluated across the MEC: using the city infrastructure for the edge deployment, and using the cloud node from the public mobile network operator (NOS[1]). The end goal is to explore how to leverage multi-site infrastructure to minimize the service-level latency for critical services. The results show the reduction of the end-to-end latency when opting to use a more hybrid approach using the edge nodes to process the messages that are received and to process the messages that are needed for notification of the vehicles, complementing with the cloud to process the more complex calculations with the collision service.

The remainder of this paper is organised as follows. Section II presents the proposed architecture. Section III details the system workflow which follows a microservices architecture, while section IV presents the different scenarios regarding the use of infrastructure federation. Finally, Section V presents and discusses the results, and Section VI enumerates the conclusions and directions for future work.

II. Architecture

Figure 1 presents the general architecture for the deployment of a microservices-based solution for ITS services considering the possibility of using edge and cloud computing, and using a city infrastructure federated with a mobile network operator. A previously monolithic application – *e.g.* in this case a safety application – is decoupled into several microservices, one for each main component of the original application, which communicates through an information aggregator/message broker in a Publisher/Subscriber (Pub/Sub) paradigm. In Pub/Sub architectures, publishers generate messages without knowing who will receive them, and subscribers express interest in specific messages without the knowledge of

[1]https://www.nos.pt

the source while the central message mediator/broker handles the communication, enforcing security and access control. Services communicate through one or more brokers. Therefore, an important part of this study is the decision of their placement. All these hops add critical delays that degrade the performance of a safety application. However, the architecture benefits from the placement of several processing and communications tasks closer to the end-users, such as: a) message processing to parse the required information (transmitter position, speed, etc); the collision detection algorithm that estimates the likelihood of a collision; and the creation of the warning notification to be sent to the users to warn if a collision is imminent.

The main challenges of this type of microservice-based solution lie in the services efficiency on using the edge computational resources, and in their deployment at locations where they are going to have better results in terms of the trade-off between computation capabilities and communication delay. During the evaluation, several variations of the distribution of the services will be considered. Firstly, the processing is deployed mainly in the smart city environment. This means that the system is not making the most use of the proximity to the cellular connection from/to vehicles or VRUs because, with the smartphone connected via 5G, the data will first travel to the 5G core infrastructure, then to the cloud of the mobile network provider, and then to the city infrastructure. All these hops are travelled without the opportunity to have information parsed and/or algorithm calculations in the nodes in the path. Thereafter, we distribute several microservices also into the MNO. This way, it is possible to take advantage of the proximity between the 5G communication and processing resources.

III. Microservices workflow

One of the most important characteristics in the deployment of these services is the fact that they shall be capable of being deployed at different locations without causing disruptions or conflicts within the system. The abstraction of microservices implies that each service shall be as self-contained and independent as possible to enable dynamic and flexible deployments [18].

In our approach the microservices are containerized, meaning that they can be meaninglessly deployed using Docker or orchestrated through Kubernetes. Kubernetes can take care of scaling the microservices on demand, manage fail-overs, and can also load balance the communications between the services. The use of Kubernetes for the services deployment is not evaluated in the context of this work.

A. Message processing service

The Message Processing service, presented in Figure 2, can be seen as the initialization process of the VRU safety application pipeline. Depending on the actor, two types of messages can be transmitted, following the ETSI format: Cooperative Awareness Message (CAM) by the vehicles, and Vulnerable Road User Awareness Message (VAM) by the VRUs. Since their content is different, two independent processes are used to decode their information. After the verification of such messages and the consequent identification of the transmitting actor, a *SimpleObjectInfo* message is created containing only the necessary information for the collision processing service, and published in the information aggregator broker.

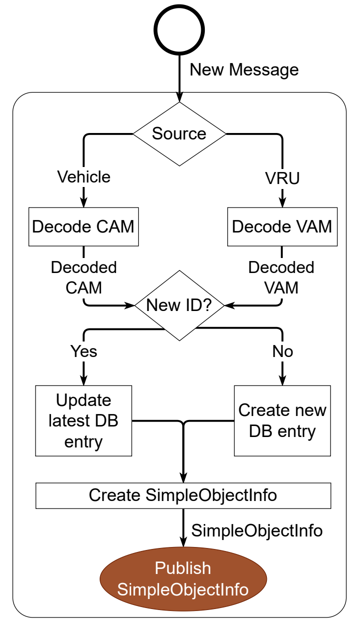


Fig. 2: Message processing flowchart.

B. Collision processing service

The collision processing service is responsible for, based on the inputs from the Message Processing service or equivalent service, compute the likelihood of a collision. It is designed as a multi-process application, allowing replication of the service and taking full advantage of parallelization. When the sources of the messages come from different entities – with one of them necessary being a VRU – the computation of the algorithm responsible for the accident detection comes into action to detect a potential collision[2].

Figure 3 presents the flowchart of the collision processing service. It starts with the reception of a *simpleObjectInfo* from a VRU, because if no VRU is detected, there is no dangerous situation to be predicted. This means that, if a *simpleObjectInfo* message from a CAM is received, the service enters in a waiting stage. Then, a *simpleObjectInfo* message from a VRU triggers the potential collision algorithm, that goes through the stages making the required verification. If a warning needs to be sent, a warning message needs to be created to notify all the users. In that case, the system sends a warning message back to the main process, which publishes it in a *warning* topic of the message broker.

[2]It is important to highlight that this work does not research on collision detection algorithms. To showcase the microservices solution, we have used the collision detection solution in [17]. Still, one can simply replace such solution with any other, assuming that the same information is used in the detection process.

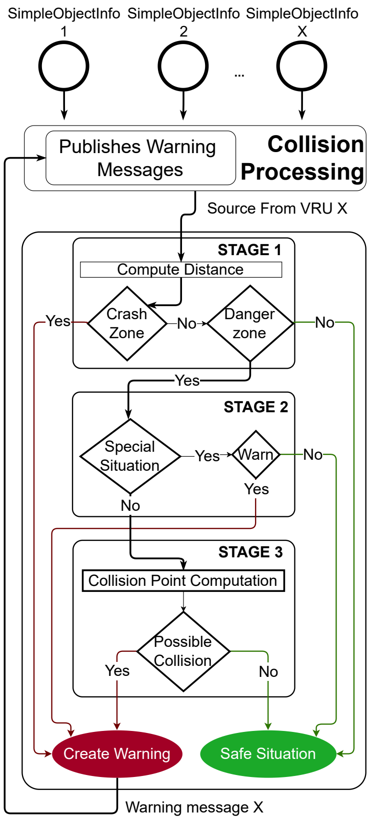


Fig. 3: Collision processing flowchart.

C. Notification relay service

The Notification Relay service is responsible for the creation and delivery of the warning message to the end users. Its main purpose is the creation and post-distribution of warning messages to end users, being them directly involved in the hazardous situation or not. Every notification of a potentially hazardous situation is transmitted using a warning message in a standardized format (ETSI Decentralized Environmental Notification Message (DENM)) to alert the other road users.

When a warning message is published, it triggers the service to compute a notification message in the DENM format, as seen in Figure 4. The warning message can be sent to multiple users, all the users, or just one. The DENM message is sent with the information of an accident or a possible accident between two possible entities, and it is broadcast so that every vehicle or VRU smartphone in range of an Road Side Unit (RSU) can receive it.

The warning message is also dispatched to the central city cloud infrastructure through a *Website DENM module*, since our solution assumes the existence of a web-based dashboard that can be used to notify a smart city operator (*e.g.* the municipality or the authorities) of all traffic occurrences such as accidents or potential accidents.

IV. Deployment Scenarios

One of the main objectives of this work is to evaluate the proposed ITS solution for VRUs awareness following

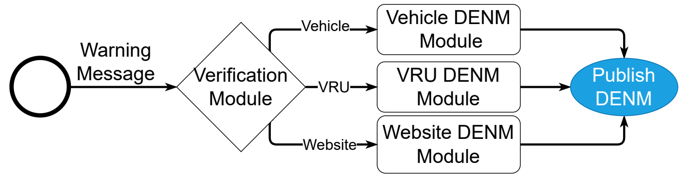


Fig. 4: Notification service flowchart.

different combinations regarding the placement of the several microservices. Before presenting the several scenarios, we describe what are the possible locations for the services deployment. As illustrated in Figure 1, these possible locations are the vehicle edge, where services can be deployed in RSUs. In the evaluation of the proposed approach, this edge is geographically located in Aveiro, Portugal, supported by the city infrastructure Aveiro Tech City Living Lab (ATCLL) [19]. Then, we have the ATCLL core, also located in Aveiro, and hosted by the city infrastructure. In addition, we have the cloud of the MNO. Thus, the scenarios considered in this work are:

1) Scenario without federation (Figure 5): solution deployed on a single infrastructure to analyse the best possible combination on the location of deployment of the services for better latency and responsiveness. This scenario considers an emulated MNO cloud, to include a delay to reach the cloud. Three variations are deployed: a new variation emerges when a microservice changes its location within the same infrastructure;
2) Scenario with federation (Figure 6): it combines the city infrastructure and the MNO infrastructure to deploy the different services. In collaboration with the MNO NOS, it was possible to use nodes in the cloud of the datacenters geographically located in Brussels (Belgium) and Madrid (Spain). The same three variations as before are also considered here.

The grey arrows and the circled numbers illustrate which microservices are offloaded to other nodes, in the different variations depicted in both Figures 5 and 6. In a first variation of the microservices placement, the main concern is to obtain the closest proximity of all them, having less number of brokers and network hops among them. It should be noted though that fewer hops do not necessarily mean a lower total latency. This depends on factors such as the underlying physical layer, the geographical proximity between the hops and the congestion of the brokers. In the second variation, with the *Notification Service* and *Message Processing* deployed in the edge node, we guarantee better proximity with the On-Board Units (OBUs) participating through ITS-G5. In the third variation, the focus is to minimize the network distance with the VRU. This last variation is better understood in the scenario with the usage of the real MNO cloud (Figure 6). With the *Collision Processing* closer to the 5G network, through the usage of the NOS infrastructure, our aim is to minimize the latency to the 5G users. Nevertheless, in the first scenario we also evaluate this case, emulating a cloud infrastructure with the inclusion of artificial delays of 50 ms

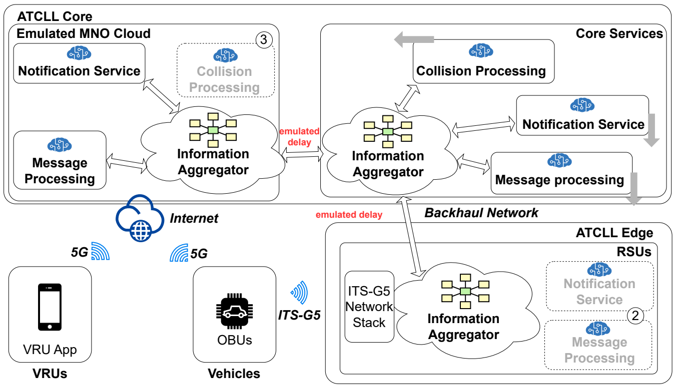


Fig. 5: Scenario without federation, using ATCLL and emulated MNO Cloud. Grey arrows indicate the offloading of the microservices from variation 1 to variations 2 and 3. The new locations are indicated by (2) and (3).

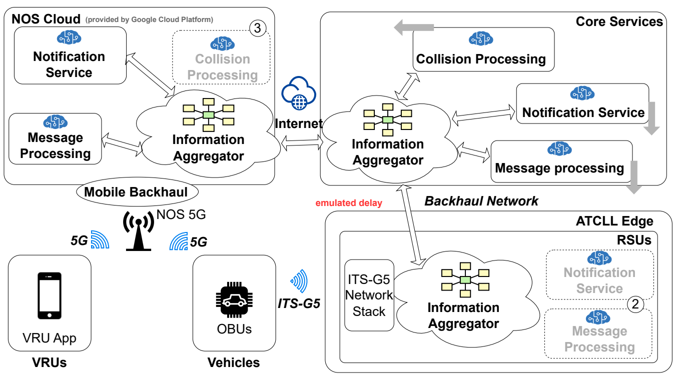


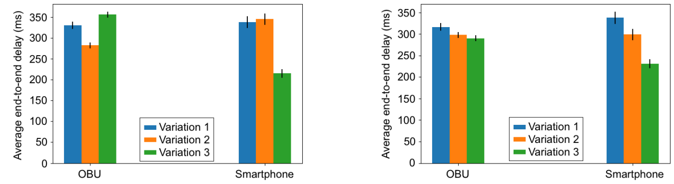
Fig. 6: Scenario with federation, using ATCLL and MNO Cloud. Grey arrows indicate the offloading of the microservices from variation 1 to variations 2 and 3. The new locations are indicated by (2) and (3).

between the nodes, through the `tc` Linux utility.

V. Evaluation

ETSI has defined that safety applications involving VRUs must be compliant with receiving messages at a sufficient data rate. In [20] it is mentioned that, for collision risk warning applications, 300 ms of maximum end-to-end latency time is required to guarantee timeliness reactions by the users. To assess if our approach complies with this target, we performed several tests, in real-world, using the ATCLL platform, one vehicle's OBU, and one VRU in potential real collision situations - defined by an extreme closeness between the vehicle and VRU and repeated 100 times. The 95% confidence interval is overlapped in the graphs.

Figure 7a illustrates the end-to-end delay, in a notification situation, of both vehicle and VRU on a side-to-side comparison without federation. The end-to-end delay of all the variations is aligned with the target defined by ETSI. Regarding the notification of the vehicle, it can be seen that, by placing the message processing and the notification services in



(a) Average end-to-end delay without federation.

(b) Average end-to-end delay with federation.

Fig. 7: Scenario without federation vs scenario with federation.

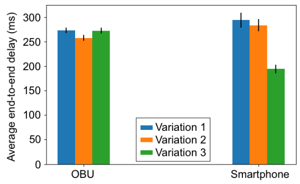
the RSUs (variation 2), we are able to reduce it almost 50 ms when compared to variation 1. On the other hand, the results also show that, by moving the collision processing service to the cellular edge (variation 3), the VRU is notified almost 100 ms faster when compared to other variations. However, such variation is the one with the highest notification delay for the vehicle.

Figure 7b shows the same three variations but now leveraging the concept of federation and using the real cloud node of the MNO. Comparing the end-to-end latency, we can observe that two of the variations are below the threshold defined by ETSI, for both the vehicle OBU side and the smartphone side. These results show even clearer that the *Collision Processing* service being closer to the 5G represents an improvement in the end-to-end latency to warn the VRU without compromising the latency to deliver the warning through ITS-G5.

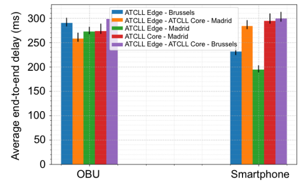
During our experiments, there was still the possibility to migrate the cloud node from a Brussels (Belgium) datacenter to Madrid (Spain), geographically closer to the tests' origin at Aveiro (Portugal). Figure 8a shows the same set of combinations using the federation with the cloud node in Madrid. The end-to-end delay to deliver the warning to the vehicle is 30 ms to 50 ms faster for all the variations, when comparing to the previous iteration (federation using the Brussels node). For the warning message to the smartphone, the improvement is in the same amount of time. For instance, for variation 3, the notification is now delivered in approximately 200 ms. With these results, we demonstrate that the microservice-based architecture for collision detection between VRUs and vehicles benefits from the distribution of the services throughout nodes that are geographically closer to the users, *i.e.* using road infrastructure edge nodes with ITS-G5 and MNO infrastructure with processing capabilities closer to the 5G network.

To better compare the different cases and their relative improvements, Figure 8b presents some of the infrastructure combinations where the end-to-end delay is below the temporal requirement of 300 ms. Once again, it shows that the variation that considers the *Notification* service and *Message Processing* service in both ATCLL edge and Madrid cloud node, and the *Collision Processing* service in Madrid (ATCLL Edge - Madrid, in the figure), is the best overall placement to deliver prompt warnings to the end users. The VRU is now notified in ≈200 ms and the vehicle is notified in ≈270 ms.

In all scenarios, the usage of a Message Queuing Telemetry



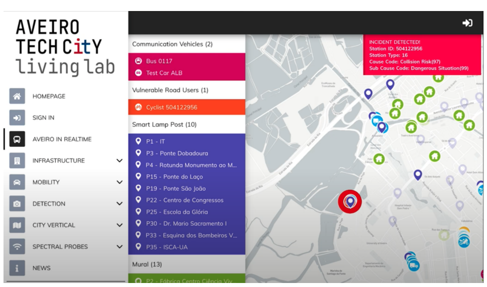
(a) Average end-to-end delay with federation using the cloud node in Madrid, Spain.



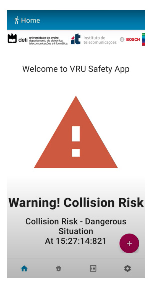
(b) Comparison of several combinations with end-to-end delay below 300 ms.

Fig. 8: Results using Madrid datacenter and comparison with the other combinations.

(a) Scenario where the users are in danger of collision.



(b) Car dashboard with notification.



(c) VRU app with notification.

Fig. 9: Functional evaluation of the system on a real scenario with an imminent collision.

Transport (MQTT) based solution for inter-service communication proved a burden for the scalability of the system, with current and future work focusing on replacing with Data Distribution System (DDS) for most scenarios.

After the temporal performance analysis, the system was demonstrated in the road, using the real infrastructure of the ATCLL, which included an RSU and an edge node processing placed in smart lampposts, and the smartphone using the 5G network from NOS. Figure 9a presents a configuration in which the user and the vehicle are in a potentially dangerous situation. When both users start to get nearby, the algorithm processes that information and assesses the danger. If there is the possibility of collision, both users are notified, as figures 9b (smartphone screenshot) and 9c (car dashboard) show. This way, the users have a notification of the possible collision and can try to avoid the incident. These user interfaces are based on the previous work presented in [17]. With this test, it is possible to prove that the system works in a real scenario, and that it can be used to prevent accidents, using this new microservices approach. A video of the system in operation is available online[3].

VI. Conclusions and Future Work

This paper proposed a scalable and reliable solution for VRU safety, specifically to attend the collision prevention safety applications, focusing on the deployment alternatives of the services associated to such use cases and their timing evaluation. This approach is based on microservices to target the always critical VRU safety use-case, in a multi-site scenario, using the federation between a road infrastructure MEC and a mobile network operator cloud. Using this approach, several scenarios and variations have been analysed to deploy the different microservices, investigating the best approach to minimize the service-level latency of the safety application. The proposed work included the definition of the several microservices and their location, the interaction between them, and their deployment in a plethora of different scenarios within the smart city ATCLL infrastructure and in the mobile network operator.

The results have shown the potential from the distribution of microservices through the edge and cloud, with a strong impact on improving the efficiency of ITS. Depending on the services' location, the latency of the VRU and vehicle's notification is deeply affected, but using a federated scenario we were able to keep the VRU's notification delay around the 200 ms, with better results being achieved if a closer mobile network provider cloud platform is used, as it was shown through the location of the MNO cloud node from Brussels to Madrid.

Future work is now being accomplished in several avenues, from the evolution to a peer-to-peer communication model through DDS, the federation where the MEC is also located in the mobile network operator, and the exploration of new safety scenarios, including immersive scenarios with AR/VR. Mechanisms for automated deployment and scaling - through Kubernetes and service migration between multi MEC sites - and the evaluation of its impact in the performed evaluation is also future work.

Acknowledgments

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